



Hospital Engineering and Management

Lecture 6: Radiology and Imaging Services

Dr. Sahaj Saxena

Department of Electrical and Instrumentation Engineering
TIET

Course Code: UBM702

Learning Objectives

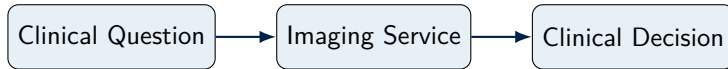
By the end of this lecture, students should be able to:

- ▶ explain the role and organization of radiology and imaging services in a hospital;
- ▶ distinguish major imaging modalities and their engineering requirements;
- ▶ describe patient flow, zoning and departmental adjacency in an imaging department;
- ▶ identify electrical, HVAC, shielding, networking and equipment-support requirements;
- ▶ explain major radiation and MRI safety principles relevant to facility operation;
- ▶ relate quality assurance, maintenance and information systems to reliable imaging services.

Connection with the Previous Lecture

Lecture 5 Established

Nursing services depend on timely diagnostic support, safe patient transfer, reliable equipment, good communication and rapid escalation when a patient deteriorates.



Today's Focus

Radiology is not only an imaging machine. It is a coordinated clinical, engineering, safety and information-management service.

Opening Scenario: Emergency Imaging at Midnight

A trauma patient arrives with suspected head and chest injuries. The emergency physician requests urgent imaging.

Clinical Needs

- ▶ rapid patient transport;
- ▶ correct imaging protocol;
- ▶ continuous monitoring;
- ▶ prompt image interpretation;
- ▶ safe return or transfer.

Engineering Needs

- ▶ scanner availability;
- ▶ uninterrupted power;
- ▶ cooling and ventilation;
- ▶ image-network connectivity;
- ▶ emergency technical support.

Diagnostic speed depends on the entire imaging system, not only on the scanner.

What Is a Radiology and Imaging Service?

Definition

A radiology and imaging service uses medical imaging technologies to support diagnosis, treatment planning, image-guided procedures and follow-up of patients.

It integrates:

- ▶ patients, radiologists, technologists, nurses and support staff;
- ▶ imaging equipment and accessories;
- ▶ radiation or magnetic-field safety controls;
- ▶ electrical power, HVAC, shielding and other infrastructure;
- ▶ digital image storage, communication and reporting systems;
- ▶ quality assurance, maintenance and regulatory compliance.

Major Imaging Modalities

Modality	Energy / Principle	Typical Strength	Important Engineering Concern
Radiography	X-rays	rapid projection imaging	shielding, detector and generator QA
CT	rotating X-ray system	fast cross-sectional imaging	high electrical and cooling demand
Fluoroscopy	continuous/pulsed X-rays	real-time guidance	dose management and room shielding
Mammography	low-energy X-rays	dedicated breast imaging	image-quality and compression-system QA
Ultrasound	acoustic waves	portable real-time imaging	probe care, electrical safety, infection control
MRI	magnetic field + RF	excellent soft-tissue contrast	access control, RF environment and magnetic safety

Ionizing and Non-Ionizing Imaging

Ionizing Radiation

- ▶ radiography;
- ▶ CT;
- ▶ fluoroscopy;
- ▶ mammography.

Requires radiation-protection measures and appropriate regulatory control.

Non-Ionizing Energy

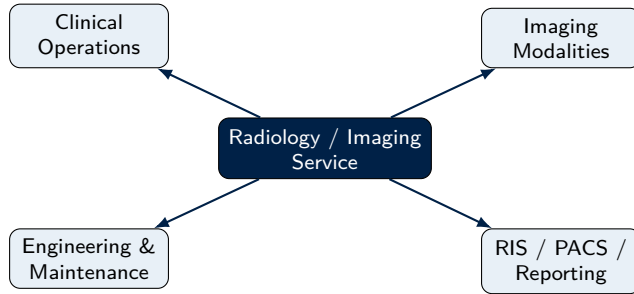
- ▶ ultrasound;
- ▶ MRI.

Different hazards remain: acoustic, electrical, thermal, magnetic and device-related risks.

Key Message

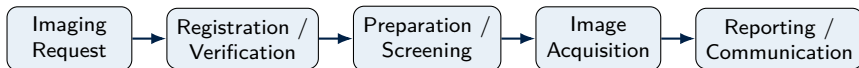
“No ionizing radiation” does not mean “no safety requirements.”

Functional Organization of an Imaging Department



- ▶ **Clinical:** referral review, patient preparation, scanning and reporting;
- ▶ **Technical:** technologists, modality operation and protocol implementation;
- ▶ **Engineering:** utilities, equipment performance, maintenance and safety support;
- ▶ **Digital:** scheduling, worklists, image storage, retrieval and report distribution.

Typical Patient Flow



Flow Must Also Accommodate

outpatients, inpatients on beds or wheelchairs, emergency patients, infection-control needs, accompanying staff, contrast preparation and post-procedure observation.

Location and Departmental Adjacency

Imaging should be planned for efficient access from high-demand clinical areas.

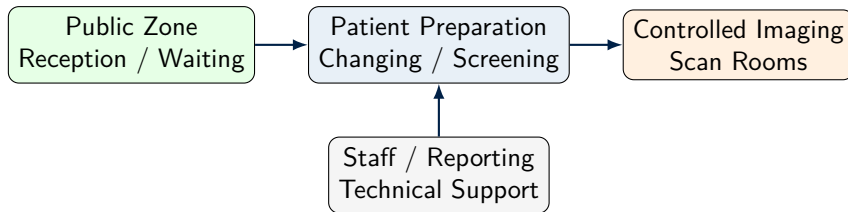
Important Adjacencies

- ▶ emergency department;
- ▶ outpatient services;
- ▶ inpatient circulation;
- ▶ ICU and operation theatre routes;
- ▶ ambulance / trauma access where relevant.

Planning Objectives

- ▶ short transfer distance;
- ▶ adequate trolley turning space;
- ▶ controlled access;
- ▶ separation from unrelated public traffic;
- ▶ convenient service access for equipment.

Functional Zoning of the Department



Design Principle

Access becomes progressively more controlled as the patient moves from public areas to imaging rooms.

Reception, Waiting and Patient Preparation

Important design features include:

- ▶ clear registration and wayfinding;
- ▶ separate consideration for ambulatory, wheelchair and trolley patients;
- ▶ patient privacy during changing and preparation;
- ▶ accessible toilets and changing cubicles;
- ▶ space for family members where appropriate;
- ▶ safe storage of personal belongings;
- ▶ screening areas for pregnancy, implants, allergies or other modality-specific risks;
- ▶ observation space for selected contrast or interventional procedures.

General Radiography Room

Typical components:

- ▶ X-ray tube and support system;
- ▶ examination table / wall stand;
- ▶ digital detector;
- ▶ operator control console;
- ▶ patient positioning accessories;
- ▶ protective barriers and warning indicators.

Planning Questions

- ▶ Can a trolley enter and turn safely?
- ▶ Is the operator protected and able to observe the patient?
- ▶ Can service personnel access the equipment?
- ▶ Are cables and floor paths safe?

Engineering Requirements for X-Ray Rooms

- ▶ room layout and shielding appropriate to equipment type, workload and surrounding occupancy;
- ▶ dedicated electrical supply as specified by the equipment manufacturer;
- ▶ protective earthing and suitable electrical distribution;
- ▶ stable environmental conditions for generator, detector and computing equipment;
- ▶ warning signs / indicators and controlled access;
- ▶ network connectivity for worklist and image transfer;
- ▶ space for acceptance testing, calibration and preventive maintenance;
- ▶ safe equipment movement and replacement route.

Important

Shielding should be designed by competent professionals using the applicable regulatory requirements; wall thickness should not be guessed from generic values.

Computed Tomography (CT)

CT uses a rotating X-ray source and detector system to generate cross-sectional images.

Clinical Advantages

- ▶ very rapid imaging;
- ▶ trauma and emergency use;
- ▶ high spatial detail;
- ▶ angiographic and multiphase studies;
- ▶ image-guided procedures in some settings.

Engineering Features

- ▶ high-power X-ray generator;
- ▶ rotating gantry;
- ▶ precision patient table;
- ▶ detector and reconstruction electronics;
- ▶ workstation and PACS interface.

CT Suite Planning

A CT suite commonly requires:

- ▶ scan room with shielding and controlled access;
- ▶ control room with direct or camera-based patient observation;
- ▶ patient preparation and changing area;
- ▶ contrast-injection support and emergency-response provisions;
- ▶ adequate space around gantry and table for resuscitation / transfer;
- ▶ electrical, HVAC and equipment-cooling provisions matched to manufacturer requirements;
- ▶ injector, monitor and accessory storage;
- ▶ reliable network connection to RIS/PACS.

Magnetic Resonance Imaging (MRI)

MRI uses strong magnetic fields, radiofrequency energy and signal processing to produce images.

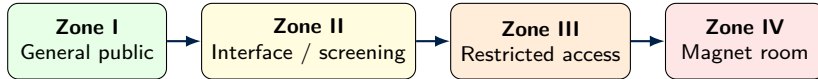
Why MRI Is Different

MRI does not use ionizing radiation, but the MR environment creates unique hazards because the magnetic field is continuously present in many systems.

Major safety concerns include:

- ▶ attraction of ferromagnetic objects;
- ▶ interaction with implants and medical devices;
- ▶ radiofrequency heating;
- ▶ acoustic noise;
- ▶ cryogen / venting issues for superconducting systems;
- ▶ restricted emergency access to the magnet room.

MRI Safety Zoning



- ▶ screening and access control become stricter toward the magnet room;
- ▶ only appropriately screened people and equipment should enter restricted MR areas;
- ▶ emergency procedures must account for the magnetic-field hazard.

MRI Engineering Requirements

- ▶ magnetic-field siting assessment and controlled access;
- ▶ RF shielding appropriate to the scanner installation;
- ▶ manufacturer-specified electrical power and grounding;
- ▶ HVAC and heat-removal capacity for scanner and equipment room;
- ▶ cryogen and emergency venting provisions where applicable;
- ▶ MR-compatible patient monitoring, anesthesia and emergency equipment when required;
- ▶ intercom, patient observation and alarm communication;
- ▶ safe delivery, installation and future magnet-replacement route.

Biomedical Engineering Point

A device may be MR Safe, MR Conditional or MR Unsafe. Unknown devices should not be casually introduced into the MR environment.

Ultrasound Services

Ultrasound creates images using high-frequency acoustic waves and is widely used because it is real-time and relatively portable.

Planning Needs

- ▶ privacy and dimmable lighting;
- ▶ examination couch and operator ergonomics;
- ▶ handwashing / infection-control access;
- ▶ network and reporting connectivity;
- ▶ adequate electrical outlets.

Engineering Focus

- ▶ probe integrity;
- ▶ image uniformity;
- ▶ electrical safety;
- ▶ battery condition for portable units;
- ▶ cleaning compatible with probe type.

Mammography Services

Mammography is a dedicated X-ray imaging service requiring high image quality and careful quality control.

Important considerations:

- ▶ patient privacy and sensitive communication;
- ▶ dedicated positioning and compression system;
- ▶ detector and image-display performance;
- ▶ radiation shielding based on installation design;
- ▶ quality-control programme for image quality and dose-related performance;
- ▶ appropriate workflow for screening, diagnostic views and reporting;
- ▶ reliable archiving and comparison with prior examinations.

Fluoroscopy and Interventional Imaging

Fluoroscopy provides real-time X-ray imaging and may support diagnostic or image-guided interventions.

Clinical Environment

- ▶ procedure table;
- ▶ contrast / medication support;
- ▶ physiological monitoring;
- ▶ sterile supplies where required;
- ▶ anesthesia or resuscitation access.

Safety Importance

Staff may remain near the patient during imaging. Room design, protective barriers, working practices and dose optimization are therefore especially important.

Planning an Interventional Imaging Suite

- ▶ sufficient space for imaging equipment, staff, anesthesia and emergency response;
- ▶ sterile / clean supplies separated from used materials;
- ▶ appropriate electrical outlets, medical gases and suction where clinically required;
- ▶ lighting that supports both image viewing and procedures;
- ▶ HVAC appropriate to procedure type and institutional infection-control requirements;
- ▶ ceiling or floor loading / support compatible with equipment configuration;
- ▶ shielding, access control and staff protective equipment;
- ▶ convenient scrub, recovery and support areas depending on service scope.

Portable and Bedside Imaging

Portable imaging brings the equipment to patients who cannot be safely transferred.

Common Locations

ICU, emergency department, operating rooms, isolation areas and inpatient wards.

Key management issues:

- ▶ safe transport and parking of mobile equipment;
- ▶ battery charging and electrical safety;
- ▶ infection-control between patients;
- ▶ radiation protection of nearby persons for mobile X-ray use;
- ▶ image identification and wireless/network connectivity;
- ▶ preventive maintenance despite frequent movement and rough handling.

RIS, PACS and Hospital Information Systems

RIS

Radiology Information System typically supports scheduling, registration, worklists, examination status and reporting workflow.

PACS

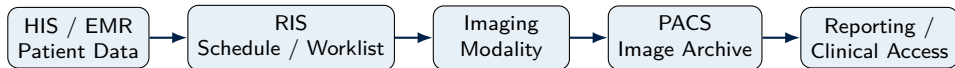
Picture Archiving and Communication System stores, retrieves, distributes and displays medical images.

Integration with hospital information systems reduces repeated data entry and improves availability of images and reports.

Engineering Relevance

Network failure can stop a technically functional scanner from delivering a clinically useful service.

Digital Imaging Data Flow



Interoperability Concept

DICOM is widely used for medical-image data exchange, while healthcare information systems may use additional interoperability standards and interfaces.

Reporting and Image-Viewing Areas

Diagnostic reporting is sensitive to the viewing environment.

- ▶ controlled ambient lighting and glare;
- ▶ appropriate diagnostic displays and workstation layout;
- ▶ ergonomic seating and monitor positioning;
- ▶ reliable network access to current and prior studies;
- ▶ voice-recognition / reporting support where used;
- ▶ acoustic privacy for communication and reporting;
- ▶ backup workflow during PACS or network downtime.

Hospital Engineering Connection

Lighting, heat load, power quality, network reliability and room acoustics influence the reporting environment.

Radiation Shielding: The Engineering Concept

Shielding design considers more than the imaging machine itself.

- ▶ radiation source and operating mode;
- ▶ expected workload and beam directions;
- ▶ distance from source to surrounding areas;
- ▶ occupancy of adjacent spaces;
- ▶ structural materials and protective barriers;
- ▶ doors, windows and penetrations;
- ▶ regulatory design criteria and verification.

Source + Workload + Geometry + Occupancy + Barrier = Shielding Design

Radiation Safety: Core Principles

Justification

Use an ionizing-radiation procedure when the expected clinical benefit justifies the exposure.

Optimization

Perform the examination so that the required clinical information is obtained with radiation exposure kept as low as reasonably achievable under the clinical circumstances.

- ▶ use appropriate protocols and collimation;
- ▶ avoid unnecessary repeat exposures;
- ▶ maintain equipment performance;
- ▶ use protective measures and good working practices;
- ▶ monitor and investigate unusual exposure events.

AERB Regulatory Context in India

For medical diagnostic X-ray equipment in India, radiation-safety regulation includes licensing and facility requirements under the applicable AERB framework.

- ▶ equipment and facility must satisfy applicable regulatory requirements;
- ▶ adequate room layout and shielding are required;
- ▶ appropriately qualified / trained personnel are part of safe operation;
- ▶ quality assurance and radiation-safety practices must be maintained;
- ▶ operation should begin only after the required regulatory authorization is obtained.

For Students

Regulatory requirements can change. Hospital engineers should always consult the current AERB requirements during actual planning or procurement.

Personnel Radiation Protection

Practical controls may include:

- ▶ controlled access to radiation areas;
- ▶ distance and positioning during exposure;
- ▶ fixed or mobile protective barriers;
- ▶ personal protective equipment appropriate to the procedure;
- ▶ personal dosimetry for designated occupationally exposed staff;
- ▶ warning signs and exposure indicators;
- ▶ training and periodic safety review;
- ▶ investigation of unusual or unintended exposures.

Quality Assurance and Quality Control

Quality Assurance (QA)

The organized programme used to ensure that the imaging service consistently produces clinically useful results safely and efficiently.

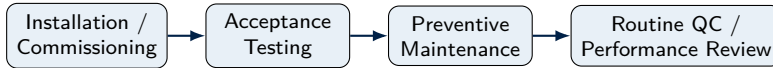
Quality Control (QC)

Technical tests and measurements used to verify equipment and image-system performance.

Examples of QC areas:

- ▶ image quality and detector performance;
- ▶ exposure / dose-related performance for X-ray systems;
- ▶ mechanical and safety interlocks;
- ▶ display and workstation performance;
- ▶ ultrasound probe and image-uniformity checks;
- ▶ MRI system and coil performance checks.

Acceptance Testing, Maintenance and Calibration



- ▶ verify performance before clinical use;
- ▶ establish baseline measurements;
- ▶ schedule manufacturer- and risk-based maintenance;
- ▶ control software updates and service changes;
- ▶ document faults, repairs, downtime and return-to-service checks.

Electrical Power and Power Quality

Imaging systems can be sensitive to electrical disturbances.

- ▶ dedicated supply and distribution as specified by the vendor;
- ▶ correct earthing / bonding arrangements;
- ▶ protection from voltage disturbances where required;
- ▶ UPS for computers, consoles, PACS/network components and controlled shutdown where appropriate;
- ▶ emergency power strategy based on clinical criticality and manufacturer limitations;
- ▶ safe cable routing and accessible isolation points;
- ▶ planned power-down and restart procedures after outages.

Important

A large scanner cannot simply be connected to a generic UPS without engineering analysis of load, inrush, harmonics and manufacturer requirements.

HVAC, Cooling and Environmental Control

Imaging equipment converts substantial electrical energy into heat.

Why HVAC Matters

- ▶ electronics reliability;
- ▶ detector stability;
- ▶ patient comfort;
- ▶ equipment-room heat removal;
- ▶ continuous service availability.

Design Inputs

- ▶ equipment heat load;
- ▶ room occupancy;
- ▶ required temperature / humidity range;
- ▶ ventilation needs;
- ▶ redundancy for critical cooling.

Infection Prevention in Imaging Areas

Imaging departments receive patients from many clinical areas, including patients with transmissible infections.

- ▶ clean high-touch surfaces between patients as required;
- ▶ use modality-compatible disinfectants;
- ▶ apply appropriate probe cleaning / high-level disinfection processes where indicated;
- ▶ manage contaminated linen and waste safely;
- ▶ coordinate portable imaging in isolation rooms;
- ▶ maintain hand-hygiene facilities and supplies;
- ▶ prevent damage to detectors, coils and probes during cleaning.

Engineering-Nursing-Technologist Interface

Cleaning procedures must control infection without damaging sensitive equipment.

Contrast and Patient-Support Requirements

Some imaging examinations require intravenous contrast, sedation or additional monitoring. Facility planning may therefore require:

- ▶ preparation and cannulation space;
- ▶ contrast storage and injector support;
- ▶ patient observation before / after selected procedures;
- ▶ emergency call and resuscitation access;
- ▶ oxygen, suction and monitoring based on service scope;
- ▶ safe transfer routes to emergency or higher-level care;
- ▶ clear communication between radiology, nursing and referring teams.

Emergency Preparedness in Imaging

Imaging areas require plans for events such as:

- ▶ patient cardiac or respiratory arrest;
- ▶ contrast reaction;
- ▶ electrical power failure;
- ▶ scanner failure with patient on the table;
- ▶ fire or smoke event;
- ▶ PACS / network outage;
- ▶ MRI projectile or implant-related safety event;
- ▶ loss of cooling or environmental control.

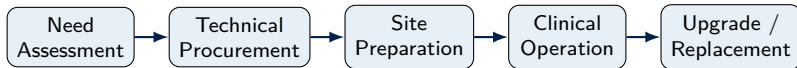
Planning Principle

Emergency response must be designed around the hazards of the modality; procedures used in a general ward may need modification inside controlled imaging areas.

Role of the Biomedical / Clinical Engineer

- ▶ participate in equipment specification and technical evaluation;
- ▶ assess utilities, space and installation prerequisites;
- ▶ coordinate site preparation and vendor installation;
- ▶ support acceptance testing and commissioning;
- ▶ manage preventive and corrective maintenance;
- ▶ review equipment alarms, failures and repeated service incidents;
- ▶ support electrical safety and accessory management;
- ▶ coordinate software, network and cybersecurity requirements with IT;
- ▶ plan upgrades, replacement and end-of-life decisions.

Equipment Life-Cycle View



Total-Cost Perspective

Purchase price is only one component. Site work, power, cooling, service contracts, software, detectors/coils, downtime and upgrades can strongly influence lifetime cost.

Operational Performance Indicators

A simple imaging-service dashboard may include:

Indicator	What It Reflects	Possible Management Response
Scanner downtime	equipment reliability	maintenance / spares / service response review
Patient waiting time	capacity and workflow	scheduling and bottleneck analysis
Repeat examination rate	image / process quality	protocol, positioning and equipment review
Report turnaround time	reporting workflow	staffing, prioritization and IT review
PACS outage time	digital reliability	redundancy and network-support review
Cancelled studies	preparation / system issues	referral, screening and patient-preparation redesign

Case Study 1: CT Scanner Down During Trauma Peak

The emergency department has several trauma patients, but the CT scanner stops because the equipment-room cooling fails.

Discuss

1. Is this an imaging-equipment failure or a facility-engineering failure?
2. What immediate clinical contingency is required?
3. Which teams should be notified?
4. What environmental parameters should be checked?
5. How could redundancy or alarm escalation reduce future downtime?

Lesson

Scanner availability depends on utilities and environmental systems that may be invisible to clinicians during normal operation.

Case Study 2: Ferromagnetic Object Near MRI

A patient arrives on a general ward trolley with an oxygen cylinder attached. Staff begin moving the trolley toward the MRI magnet room.

Immediate Questions

- ▶ Has the patient been screened?
- ▶ Is the trolley MR compatible?
- ▶ Is the cylinder permitted in the MR environment?
- ▶ Who controls access to the restricted area?

System Lesson

MRI safety depends on zoning, screening, labeling, staff training and access control – not on warning signs alone.

Classroom Activity: Design a Small Imaging Department

A 150-bed hospital plans to install:

- ▶ one general radiography room;
- ▶ one CT scanner;
- ▶ two ultrasound rooms.

In groups, sketch a simple block layout and identify:

1. reception and waiting area;
2. patient preparation / changing;
3. modality rooms and control areas;
4. staff / reporting area;
5. equipment and service access;
6. two major safety controls;
7. three essential engineering utilities.

Mini Project Ideas

- ▶ patient-flow and waiting-time study in a radiology department;
- ▶ imaging-equipment downtime and root-cause dashboard;
- ▶ CT/MRI utility-readiness checklist for procurement;
- ▶ PACS downtime contingency workflow;
- ▶ portable X-ray infection-control and battery-readiness audit;
- ▶ radiology preventive-maintenance scheduling model;
- ▶ imaging-room risk map covering power, HVAC, safety and network failure;
- ▶ lifecycle-cost comparison of two imaging-equipment procurement options.

Expected Approach

Define the problem, map the workflow, identify measurable indicators and propose an engineering-management improvement.

Key Takeaways

- ▶ Radiology combines clinical workflow, sophisticated equipment, infrastructure, safety and digital information systems.
- ▶ Different modalities require different room layouts, utilities and hazard controls.
- ▶ X-ray and CT planning requires radiation-protection design and applicable regulatory authorization.
- ▶ MRI requires strict screening, zoning and control of objects and devices entering the magnetic environment.
- ▶ PACS, RIS and network services are essential parts of modern imaging operations.
- ▶ QA, QC and planned maintenance support image quality, patient safety and equipment availability.
- ▶ Biomedical engineers contribute from procurement and site planning through maintenance and replacement.

A safe image is produced by a complete system – not by a scanner alone.

Assignment for the Next Lecture

Study the radiology / imaging services of one hospital through a visit, website or published information.

Prepare a one-page note containing:

1. imaging modalities available;
2. a simple patient-flow diagram;
3. three major engineering requirements of one selected modality;
4. two safety controls;
5. one digital system used or expected (RIS/PACS/HIS);
6. one possible equipment or infrastructure failure;
7. one improvement suggestion from a biomedical-engineering perspective.

Submission Format

One page, individually prepared, with one small block diagram or departmental layout.

Suggested Reading and Safety References

- ▶ G. D. Kunders, *Hospitals: Facilities Planning and Management*, Tata McGraw-Hill.
- ▶ B. M. Sakharkar, *Principles of Hospital Administration and Planning*, Jaypee Brothers Medical Publishers.
- ▶ Atomic Energy Regulatory Board (AERB), *Radiation Safety in Manufacture, Supply and Use of Medical Diagnostic X-Ray Equipment*, AERB/RF-MED/SC-3 (Rev. 2).
- ▶ International Atomic Energy Agency (IAEA), *Radiation Protection and Safety in Medical Uses of Ionizing Radiation*.
- ▶ IAEA, *Handbook of Basic Quality Control Tests for Diagnostic Radiology*.
- ▶ American College of Radiology, *Manual on MR Safety*, 2026 edition.

Next Lecture: Clinical Laboratory Services

Thank You

Questions and Discussion